Solving the Minimum Variance Portfolio Fast: A Krylov Perspective

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Abstract

We revisit the long-only minimum variance portfolio problem and reframe it as a linear algebra problem amenable to Krylov subspace methods. Rather than forming the sample covariance matrix explicitly, we operate directly on the returns matrix using matrix-vector products. This yields a family of solvers — direct KKT, MINRES, and constraint-eliminated CG — that expose a clean hierarchy of accuracy, speed, and scalability. We draw an explicit connection between the shrinkage estimator of Ledoit and Wolf and preconditioning, unifying two literatures that rarely speak to each other. Block extensions of MINRES, well suited to rolling-window or multi-universe batch problems, are identified as a natural next step.

1 Introduction

The minimum variance portfolio is the canonical starting point for systematic equity allocation. Given a matrix of asset returns, the goal is to find the portfolio with the smallest possible variance — a well-posed convex optimisation problem with a classical solution via Markowitz (1952).

In practice, the problem is almost always solved by forming the sample covariance matrix and passing it to a general-purpose QP solver. This works, but obscures the structure of the problem. The covariance matrix is never observed directly — it is estimated from a returns matrix — and the solver has no knowledge of this origin.

We argue that working directly with the returns matrix is both numerically sounder and algorithmically richer. The resulting formulations connect naturally to Krylov subspace methods and to shrinkage estimation. The closest related work applies GMRES to the MVP problem when the covariance matrix is singular [1]; our contribution is to avoid forming the covariance matrix altogether and to connect the shrinkage estimator of Ledoit and Wolf [4] explicitly to preconditioning.

A note on origins. The KKT system derived below is symmetric indefinite — the canonical setting for MINRES and SYMMLQ [6]. The author's 2004 diploma thesis [7], supervised by Martin H. Gutknecht (ETH Zürich), studied block extensions of precisely these two methods, and the connection to the present work is what motivates the Krylov framing adopted here.

2 Problem Formulation

Let $X \in \mathbb{R}^{T\times n}$ be the matrix of daily returns for n assets over T trading days. The long-only minimum variance problem is:

$$\min_{w} \; w^\top \Sigma w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0$$

*The first author's 2004 diploma thesis at the University of Kaiserslautern [7], supervised by Martin H. Gutknecht (ETH Zürich), developed block extensions of MINRES and SYMMLQ. The present work revisits those same methods in a new setting twenty-two years later.

where $\Sigma = \frac{1}{T} X^\top X$ is the sample covariance matrix. Since $\Sigma \propto X^\top X$, this is equivalent to:

$$\min_w \|Xw\|^2 \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0$$

This reframing is key: the objective is now a squared norm of Xw, and we can evaluate $w \mapsto X^\top(Xw)$ using two matrix-vector products, without ever forming $X^\top X$.

3 The KKT System

Ignoring the inequality constraints for now, the Lagrangian for the equality-constrained problem is:

$$\mathcal{L}(w, \lambda) = \|Xw\|^2 - \lambda(\mathbf{1}^\top w - 1)$$

Setting derivatives to zero gives the stationarity condition $2X^\top Xw = \lambda \mathbf{1}$. Combined with the constraint $\mathbf{1}^\top w = 1$, this yields the augmented KKT system:

$$\begin{pmatrix} 2X^\top X & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix} \begin{pmatrix} w \\ \lambda \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}$$

This $(n+1) \times (n+1)$ system is **symmetric but indefinite** — the zero in the bottom-right corner ensures the matrix has both positive and negative eigenvalues. This rules out standard CG, but it opens the door to MINRES.

The long-only constraint $w \geq 0$ is handled via an **active set method**: solve the KKT system on the current active set, drop any asset with a negative weight, and repeat. The process terminates in at most n iterations, each requiring one solve.

4 Krylov Solvers for the Indefinite System

4.1 CVXPY Reference Implementation

Before developing the Krylov solvers we establish a reference solution using CVXPY [2], a domain-specific language for convex optimisation. The minimum variance problem maps directly onto a least-squares objective, which CVXPY dispatches to a general-purpose interior-point solver.

```
import cvxpy as cp

n = R.shape[1]
w = cp.Variable(n)
problem = cp.Problem(
    cp.Minimize(cp.sum_squares(R @ w)),
    [cp.sum(w) == 1, w >= 0]
)
problem.solve()
weights = w.value
```

The formulation is transparent and correct, and CVXPY handles the long-only constraint natively. Its cost, however, is substantial: interior-point methods require $O(n^3)$ work per iteration, and the overhead of problem construction and solver communication adds further latency. At $n = 1000$ this amounts to nearly 5 seconds (Table 1), making CVXPY unsuitable for production use but ideal as a ground-truth reference.

4.2 Direct Solve (Baseline)

For small n, `numpy.linalg.solve` applied to the explicit KKT matrix is fast and exact. It requires forming $X^\top X$ — an $O(n^2T)$ operation — but for $n \ll T$ this is not the bottleneck.

```
def build_kkt(R):
    """Build the KKT system matrix and RHS for the
    minimum variance problem.

    Constructs the (N+1) x (N+1) indefinite KKT system for:

        min  w^T (R^T R) w
        s.t. sum(w) = 1

    The system has the form:

        [ 2 R^T R   1 ] [ w ]   [ 0 ]
        [ 1^T       0 ] [lam] = [ 1 ]
    """
    n = R.shape[1]
    A = np.zeros((n + 1, n + 1))
    A[:n, :n] = 2 * R.T @ R          # form X^T X explicitly
    A[:n, n] = 1                     # budget constraint column
    A[n, :n] = 1                     # budget constraint row
    b = np.zeros(n + 1);  b[n] = 1
    return A, b

active = np.ones(n, dtype=bool)
while True:
    A, b = build_kkt(R[:, active])
    sol = np.linalg.solve(A, b)
    w_a = sol[:active.sum()]
    if np.all(w_a >= 0):
        break
    active[np.where(active)[0][w_a < 0]] = False
```

4.3 MINRES

MINRES [6] is a Krylov method for symmetric indefinite systems. It minimises the residual $\|Ax - b\|$ over the Krylov subspace $\mathcal{K}_k(A, b)$. Applied to our KKT system, the key ingredient is a matrix-free `LinearOperator` that applies the KKT matrix via two matrix-vector products on the returns matrix — $X^\top X$ is never formed.

```
from scipy.sparse.linalg import LinearOperator, minres

def matvec(x):
    out = np.empty(n + 1)
    out[:n] = 2.0 * (R.T @ (R @ x[:n])) + x[n]
    out[n]  = x[:n].sum()
    return out

kkt  = LinearOperator(shape=(n+1, n+1), matvec=matvec)
b    = np.zeros(n + 1);  b[n] = 1.0
w, _ = minres(kkt, b)
```

Convergence depends on the eigenvalue distribution of the KKT matrix. For well-conditioned problems MINRES converges in very few iterations; Section 5 discusses how Ledoit-Wolf shrinkage

compresses the spectrum and accelerates convergence in the ill-conditioned case.

4.4 Constraint-Eliminated CG

A cleaner approach avoids the indefinite system entirely. We parameterise the solution as:

$$w = w_0 + Pv$$

where $w_0 = \frac{1}{n}\mathbf{1}$ is any particular solution satisfying $\mathbf{1}^\top w_0 = 1$, and $P \in \mathbb{R}^{n\times(n-1)}$ is an orthonormal basis for the null space of $\mathbf{1}^\top$, obtained via QR factorisation of $\mathbf{1}$.

Any choice of v gives a feasible w. The reduced unconstrained problem in v is:

$$\min_v \|Xw_0 + XPv\|^2$$

with normal equations:

$$(XP)^\top(XP)\,v = -(XP)^\top(Xw_0)$$

The matrix $(XP)^\top(XP)$ is **symmetric positive definite**, so standard CG applies. The matrix-vector product $v \mapsto P^\top X^\top(X(Pv))$ requires four cheap operations and never forms $X^\top X$ or $(XP)^\top(XP)$ explicitly. In the implementation P is never stored: it is applied via an implicit Householder reflector in $O(n)$ time.

```
from scipy.sparse.linalg import LinearOperator, cg

# Implicit null-space basis via Householder reflector
sqrt_n = np.sqrt(float(n))
beta   = 1.0 / (n + sqrt_n)
v0     = 1.0 + sqrt_n

def P_apply(y):                        # P y  (n-1 -> n)
    s = y.sum()
    out = np.empty(n);
    out[0] = -beta * v0 * s;
    out[1:] = y - beta * s
    return out

def PT_apply(u):                       # P^T u  (n -> n-1)
    s = v0 * u[0] + u[1:].sum()
    return u[1:] - beta * s

w0  = np.ones(n) / n                   # particular solution: 1^T w0 = 1
rhs = -PT_apply(R.T @ (R @ w0))

def matvec(y):
    return PT_apply(R.T @ (R @ P_apply(y)))

A_red = LinearOperator(shape=(n-1, n-1), matvec=matvec)
v, _  = cg(A_red, rhs)
w     = w0 + P_apply(v)                # feasible: 1^T w = 1
```

5 Ledoit-Wolf Shrinkage

The sample covariance matrix $\frac{1}{T}X^\top X$ is a poor estimator when n is large relative to T: its smallest eigenvalues are severely downward-biased, and the eigenvalue spread — the ratio of largest to smallest — far exceeds that of the true covariance. Ledoit and Wolf [4] propose

replacing the sample covariance with a convex combination of the sample estimate and the identity:

$$\Sigma_{\mathrm{LW}} = (1-\alpha)\cdot\frac{X^\top X}{T} + \alpha I$$

The shrinkage intensity $\alpha \in (0,1)$ is chosen analytically to minimise the expected Frobenius distance to the true covariance, requiring no cross-validation. The effect on the spectrum is direct: adding αI lifts every eigenvalue by α, bounding the smallest away from zero and compressing the overall spread. The condition number satisfies:

$$\kappa(\Sigma_{\mathrm{LW}}) \le \frac{\lambda_{\max}+\alpha}{\alpha}$$

and is fully controlled by α. Subsequent work by the same authors provides an analytical nonlinear shrinkage formula [5] that applies a different shrinkage coefficient to each sample eigenvalue, achieving a strictly smaller Frobenius loss when n/T is not negligible.

In the implementation, shrinkage is applied by augmenting the return matrix rather than forming Σ_{LW} explicitly. Setting $c = 1-\alpha$ and $\gamma = \alpha\cdot\|\Sigma\|_F/n$, the stacked matrix $\tilde{X} = [\sqrt{c}\,X;\ \sqrt{\gamma}\,I]$ satisfies $\tilde{X}^\top\tilde{X} = c\,X^\top X + \gamma I \propto \Sigma_{\mathrm{LW}}$, so any of the solvers above can be applied to $\tilde{X}$ without modification.

From the perspective of numerical linear algebra, compressing the eigenvalue spectrum of the system matrix is precisely what a preconditioner does. Ledoit-Wolf shrinkage therefore plays a dual role: it reduces statistical estimation error and, as a side effect, accelerates iterative solvers by clustering the eigenvalues. The two motivations are independent but arrive at the same modified matrix.

6 Numerical Results

Table 1 reports timings and iteration counts for all four solvers on a synthetic problem with $n = 1000$ assets and $T = 2000$ daily returns, run both without and with Ledoit-Wolf shrinkage.

Method	$\|Xw\|$	Time (s)	Iterations	Speedup
Without Ledoit-Wolf shrinkage				
cvxpy	0.9924	6.7192	–	1.0x
KKT direct	0.9931	0.0456	–	147.4x
MINRES	0.9929	0.0114	39	590.1x
CG (constraint-eliminated)	0.9931	0.0149	80	451.7x
With Ledoit-Wolf shrinkage				
cvxpy	1.0582	4.6214	–	1.0x
KKT direct	1.0582	0.0473	–	97.6x
MINRES	1.0583	0.0072	16	643.9x
CG (constraint-eliminated)	1.0582	0.0083	26	558.3x

Table 1: Results on 1000 assets, 2000 trading days. Speedup relative to `cvxpy` within each panel.

The impact of Ledoit-Wolf shrinkage on the Krylov solvers is immediate: MINRES iterations drop from 39 to 16 and CG from 80 to 26, reflecting the compressed eigenvalue spectrum of the regularised system. The KKT direct solve is unaffected — it does not iterate — and its runtime stays flat across both panels because forming $X^\top X$ dominates regardless of conditioning.

MINRES is the fastest solver in both settings. At $n = 1000$ it is roughly $4\times$ faster than KKT direct because it avoids the $O(n^2T)$ cost of assembling $X^\top X$ explicitly. The constraint-eliminated

CG is close behind and is the natural choice when X is sparse or when memory for $X^\top X$ is unavailable.

7 Conclusion

We have shown that the long-only minimum variance portfolio problem, when formulated directly in terms of the returns matrix, is a structured linear algebra problem that admits a clean hierarchy of solvers. Avoiding the explicit formation of the sample covariance matrix is both numerically advantageous and algorithmically revealing: it exposes the KKT system's symmetric indefinite structure and opens the door to Krylov methods that scale to large universes.

The key empirical finding is that MINRES, applied matrix-free to the KKT system, is faster than a direct solve by a factor of roughly four at $n = 1000$ and the gap widens as n grows. Ledoit-Wolf shrinkage [4] compresses the eigenvalue spectrum of the system, reducing MINRES iterations from 39 to 16 and CG iterations from 80 to 26 in our experiments. The shrinkage intensity is chosen to minimise estimation error — the acceleration of the iterative solver is a consequence, not a design goal.

The paper also draws a connection that the two literatures involved — numerical linear algebra and high-dimensional statistics — do not usually make explicit. The shrinkage estimator of Ledoit and Wolf and the preconditioning theory of Golub and Van Loan [3] address different problems but act on the same object: the eigenvalue spectrum of the system matrix. Recognising this duality suggests that advances in one field transfer directly to the other.

References

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